1. Produce summary statistics of “residual.sugar” and use its median to divide the data into two groups A and B. We want to test if “density” in Group A and Group B has the same population mean. Please answer the following questions.

```
summary(red$residual.sugar)
median_sugar <- median(red$residual.sugar)
median_sugar
```

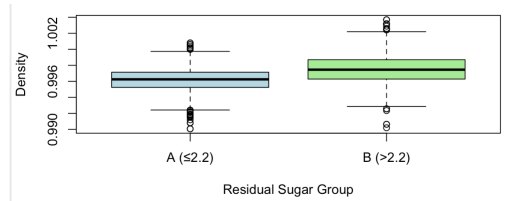
median_sugar = 2.2

- State the null hypothesis;

The mean density of wines in Group A (residual sugar ≤ 2.2) is equal to the mean density of wines in Group B (residual sugar > 2.2).

- Use visualization tools to inspect the hypothesis. Do you think the hypothesis is right or not?

```
red$group <- ifelse(red$residual.sugar <= 2.2, "A (≤2.2)", "B (>2.2)")
boxplot(density ~ group, data = red,
        xlab = "Residual Sugar Group",
        ylab = "Density",
        col = c("lightblue", "lightgreen"))
```



No, based on my visualization I do not think my hypothesis is correct.

- What test are you going to use?

I am going to use a two-sample t-test to compare the mean density between the two groups of wines.

```
t.test(density ~ group, data = red)
```

- What is the p-value?

The p-value is < 2.2e-16.

```
        Welch Two Sample t-test

data:  density by group
t = -14.697, df = 1365.2, p-value < 2.2e-16
alternative hypothesis: true difference in means between group A (≤2.2) and group B (>2.2) is not equal to 0
95 percent confidence interval:
 -0.001513022 -0.001156687
sample estimates:
mean in group A (≤2.2) mean in group B (>2.2)
             0.9961490              0.9974838

>
```

- What is your conclusion?

I reject the null hypothesis because the p-value is small. This means there is a difference between the mean density between the two groups.

- Does your conclusion imply that there is an association between “density” and “residual.sugar”?

Yes, this does imply that there is an association between “density” and “residual.sugar”.

2. Produce summary statistics of “residual.sugar” and use its 1st, 2nd, and 3rd quantiles to divide the data into four groups A, B, C, and D. We want to test if “density” in the four groups has the same population mean. Please answer the following questions.
   - State the null hypothesis;

H0: The mean density is the same across all four residual sugar groups.

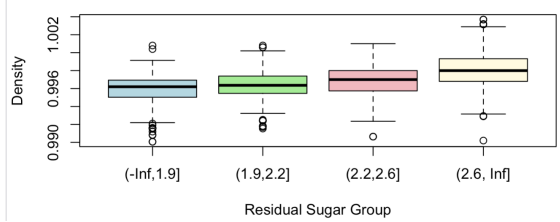
H1: At least one group’s mean is different.

   - Use visualization tools to inspect the hypothesis. Do you think the hypothesis is right or not?

```r
quantiles <- quantile(red$residual.sugar, probs = c(0.25, 0.5, 0.75))

red$group4 <- cut(red$residual.sugar,
                  breaks = c(-Inf, quantiles, Inf))

boxplot(density ~ group4, data = red,
        xlab = "Residual Sugar Group",
        ylab = "Density",
        col = c("lightblue", "lightgreen", "lightpink", "lightyellow"))
```



No, based on my visualization I do not think my hypothesis is correct.

   - What test are you going to use?

I am going to use ANOVA because there are more than two groups.

```r
anova_result <- aov(density ~ group4, data = red)
summary(anova_result)
```

   - What is the p-value?

The p-value is < 2.2e-16.

```
              Df   Sum Sq   Mean Sq F value Pr(>F)
group4         3 0.000996 0.0003321   112.8 <2e-16 ***
Residuals   1595 0.004696 0.0000029
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- What is your conclusion?

I reject the null hypothesis because the p-value is small. This means there is a difference between the mean density among the four groups.

- Does your conclusion imply that there is an association between "density" and "residual.sugar"? Compare your result here with that in Question 1. Do you think increasing the number of groups help identify the association? Would you consider dividing the data into 10 groups so as to help the discovery of the association? Why?

Yes, this does imply that there is an association between "density" and "residual.sugar". Increasing the groups from 2 to 4 helped show more variation between sugar groups. I would not consider dividing the groups into ten because each group would have too little observations.

3. Create a 2 by 4 contingency table using the categories A, B, C, D of "residual.sugar" and the binary variable "excellent" you created in Part B. Note that you have two factors: the categorical levels of "residual.sugar" (A, B, C and D) and an indicator of excellent wines (yes or no).
   - Use the Chi-square test to test if these two factors are correlated or not;

```r
red$excellent <- ifelse(red$quality >= 7, 1, 0)
tbl <- table(red$group4, red$excellent)
chi <- chisq.test(tbl)
chi
```

```
        Pearson's Chi-squared test

data:  tbl
X-squared = 5.5, df = 3, p-value = 0.1386
```

There is no significant relationship between these two factors.

- Use the permutation test to do the same and compare the result to that in (a);

```r
set.seed(123)
B <- 10000

chi_obs <- chisq.test(tbl)$statistic
perm_stats <- replicate(B, {
  tbl_perm <- table(red$group4, sample(red$excellent))
  chisq.test(tbl_perm)$statistic
})
p_perm <- mean(perm_stats >= chi_obs)
```

```
> p_perm
[1] 0.1367
```

The permutation p-value = 0.1367. It is very close to the Chi-square test, but because of this, it just means there is no significant relationship.

- Can you conclude that "residual.sugar" is a significant factor contributing to the excellence of wine? Why?

No, residual sugar is not a significant factor for wine excellence because the p-value is greater than 0.05.